

# The category of representations of the symmetric group $S_t$ , when $t$ is not a natural integer

P. Deligne

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**Abstract.** Let  $k$  be a field of characteristic 0. For  $t \in k$  which is not an integer  $\geq 0$ , we construct a semi-simple tensor category  $\text{Rep}(S_t)$  which “interpolates” the categories of linear representations (over  $k$ ) of the symmetric groups  $S_n$ , for  $n$  large. When  $t$  is a natural integer,  $\text{Rep}(S_t)$  can be “specialized” to a tensor category in at least two ways: as the usual category of representations of  $S_t$ , or as a non-semisimple tensor category.

The theory reproduces, in a simpler form, known analogous results for the linear or orthogonal groups, which will be recalled.

## Introduction

If  $G$  is an affine group scheme over a field  $k$ , the category  $\text{Rep}(G)$  of finite-dimensional linear representations of  $G$  over  $k$  is an abelian category. It is endowed with the tensor-product functor for representations

$$\otimes : \text{Rep}(G) \times \text{Rep}(G) \longrightarrow \text{Rep}(G).$$

This functor is ACU, i.e. it is endowed with associativity and commutativity constraints, and it has a unit object  $\mathbf{1}$ . It is rigid: each object  $X$  has a dual  $X^\vee$ , an object endowed with

$$\delta : \mathbf{1} \longrightarrow X \otimes X^\vee, \quad \text{ev} : X^\vee \otimes X \longrightarrow \mathbf{1}$$

such that the composite morphisms

$$X \xrightarrow{\delta \otimes X} X \otimes X^\vee \otimes X \xrightarrow{X \otimes \text{ev}} X \quad X^\vee \xrightarrow{X^\vee \otimes \delta} X^\vee \otimes X \otimes X^\vee \xrightarrow{\text{ev} \otimes X^\vee} X^\vee$$

are identities. Finally,

$$k \xrightarrow{\sim} \text{End}(\mathbf{1}).$$

In other words, in the terminology of (Deligne 1990), 2.1,  $\text{Rep}(G)$  is a tensor category over  $k$ .

If  $k$  is algebraically closed, the tensor category  $\text{Rep}(G)$  determines  $G$  up to isomorphism (Saavedra 1972). From this point of view, tensor categories generalize affine group schemes. There are examples of tensor categories other than the  $\text{Rep}(G)$ , but one has no idea how to classify them all, not even those which are semisimple and finitely  $\otimes$ -generated (and hence analogous to categories  $\text{Rep}(G)$  for  $G$  an algebraic group which is an extension of a finite group by a reductive group).

Another example of a tensor category is the category of super-representations of an affine super group scheme  $G$ . A more natural variant is to consider  $G$  endowed with  $\varepsilon : \mu_2 \rightarrow G$  such that the inner action of  $\mu_2$  on the affine algebra  $\mathcal{O}(G)$  of  $G$  defines the modulo 2 grading of  $\mathcal{O}(G)$ , and the category  $\text{Rep}(G, \varepsilon)$  of super-representations  $V$  for which the action of  $\mu_2$  gives the modulo 2 grading of  $V$ .

In characteristic  $p$ , S. Gelfand and D. Kazhdan (1992) constructed new semisimple tensor categories as “subquotients” of representation categories. By contrast, over an algebraically closed field  $k$  of characteristic 0, the only tensor categories that I know are obtained by “interpolation” of categories  $\text{Rep}(G, \varepsilon)$ .

Here is a method of interpolation. Start from an infinite family of tensor categories  $C_i$  over  $k$ , from a non-trivial ultrafilter  $F$  on the index set  $I$ , and take the ultraproduct  $C$  of the  $C_i$ . An object of  $C$  is therefore a family  $(X_i)$  of objects of the  $C_i$ , a morphism  $(X_i) \rightarrow (Y_i)$  is a germ, with respect to  $F$ , of families of morphisms  $X_i \rightarrow Y_i$ , and the tensor product is given by

$$(X_i) \otimes (Y_i) = (X_i \otimes Y_i).$$

The category  $C$  is tensorial over the field  $k_F$ , the ultrapower of  $k$ . It is not very friendly, since, even if the  $C_i$  are categories of representations of linear algebraic groups, the Hom-groups in  $C$  will be infinite-dimensional. To do better, one can start from tensor categories  $C_i$  over  $k$ , each endowed with an object  $X_i$ , and consider the full tensor subcategory  $C'$  of  $C$  generated by  $X = (X_i)$ . The objects of  $C'$  are then the subquotients of sums of

$$T^{r,s}(X) := X^{\otimes r} \otimes (X^\vee)^{\otimes s}.$$

Assume that

(\*) for all  $r$  and  $s$ , the object  $T^{r,s}(X_i)$  of  $C_i$  has finite length, bounded independently of  $i$ .

Under this hypothesis the objects of  $C'$  have finite length, and the Hom-groups are therefore finite-dimensional over  $k_F$  (consider  $\text{Hom}(X, Y)$  and apply 3.2.4).

The definition by ultraproduct is somewhat theological. In good cases, one can associate with each  $i$  a parameter  $\lambda_i$  in  $k$  describing  $(C_i, X_i)$ , and  $C'$  can be constructed purely in terms of  $\lambda = (\lambda_i) \in k_F$ . More precisely, for  $k_1$  an extension of  $k$  and  $\lambda_1 \in k_1$  transcendental over  $k$ , one can construct a tensor category  $C(\lambda_i)$  over  $k_1$ , which gives  $C'$  when one takes  $k_1 = k_F$  and  $\lambda_1 = (\lambda_i)$ . The role of ultraproducts is then only to have drawn attention to condition (\*).

The case where  $\lambda_1$  is transcendental is the easy case. More interesting, and more difficult, is to take  $\lambda_1$  arbitrary in an extension  $k_1$  of  $k$ , or at least to understand which values of  $\lambda_1$  are singular, and how.

From now on  $k$  is assumed to have characteristic 0. The case treated in this article is:

$I$  : the set of integers  $\geq 0$ ;  
 $C_n$  : the category of representations of  $S_n$ ;  
 $X_n$  : the permutation representation on  $k^n$ ;  
parameter:  $n$ ;  
singular values: the integers  $\geq 0$ .

The following cases are essentially known and will be recalled:

$I =$  the set of integers  $\geq 0$ ;  
 $C_n = \text{Rep}(\text{GL}(n))$  (resp.  $\text{Rep}(O(n))$ );  
 $X_n =$  the evident representation;  
parameter:  $n$ ;  
singular values:  $\mathbb{Z}$ .

When  $C_i$  is the category of representations of a finite group  $G_i$ , and  $X_i$  is the permutation representation defined by an action of  $G_i$  on a finite set  $B_i$ , condition (\*) is equivalent to

(\*)' for every  $r$ , the number of orbits of  $G_i$  in  $B_i^r$  is bounded independently of  $i$ .

Here are examples in which condition (\*)' is satisfied. In each of them one fixes a finite field  $\mathbb{F}_q$ , and  $I$  is the set of integers  $\geq 0$ :

- (A)  $G_n = \text{GL}(n, \mathbb{F}_q)$ ,  $B_n = \mathbb{F}_q^n$ ;
- (B)  $G_n = O(n, \mathbb{F}_q)$ ,  $B_n = \mathbb{F}_q^n$ ;
- (C)  $G_n = \text{Sp}(2n, \mathbb{F}_q)$ ,  $B_n = \mathbb{F}_q^{2n}$ .

In case (A), one can take the parameter  $\lambda_n := q^n$ , and I conjecture that the singular values of the parameter are the  $q^n$ , for  $n \geq 0$ .

I thank the referee for a careful reading, which led me to correct several obscure points.

## 1. Terminology and notation

**1.1.** We shall use the following notation.

$k$ : a field of characteristic 0, fixed throughout. For  $I$  a finite set:

$$\begin{aligned} S_I &: \text{ the symmetric group of permutations of } I; \\ k^I &: \text{ the } k\text{-vector space freely generated by } I; \\ (e_i)_{i \in I} &: \text{ its evident basis.} \end{aligned}$$

If  $I = \{1, \dots, n\}$ , we write  $S_n$  (resp.  $k^n$ ) instead of  $S_I$  (resp.  $k^I$ ).

**1.2.** Let  $(\cdot, \cdot)$  be the symmetric bilinear form on  $k^I$  defined by  $(e_i, e_j) = \delta_{ij}$ . It identifies  $k^I$  with its dual. Viewing  $k^I$  as the space of functions from  $I$  to  $k$ , via

$$u \mapsto \sum u(i)e_i,$$

one defines an algebra structure on  $k^I$  by

$$(u \cdot v)(i) = u(i)v(i).$$

For this algebra structure,

$$(u, v) = \text{Tr}(uv).$$

**1.3.** The functor  $I \mapsto k^I$ , being a left adjoint, preserves inductive limits, and in particular carries disjoint sums to direct sums. It carries finite products to tensor products, by

$$\bigotimes_{j \in J} k^{I_j} \xrightarrow{\sim} k^{\prod I_j}, \quad \bigotimes e_{a(j)} \mapsto e_{(a(j))_{j \in J}}.$$

**1.4.** If a group  $G$  acts on  $I$ , it acts on  $k^I$  by transport of structure:

$$ge_i = e_{gi}.$$

The linear representation  $k^I$  of  $G$  is called the permutation representation defined by the  $G$ -set  $I$ . The form  $(\cdot, \cdot)$  and the algebra structure of 1.2 are  $G$ -invariant. The functor  $I \mapsto k^I$  still carries disjoint sums (resp. finite products) of  $G$ -sets to direct sums (resp. tensor products) of representations.

**1.5.** Let  $(U_j)_{j \in J}$  be a finite family of finite sets. A gluing of the  $U_j$  is a finite set  $A$  endowed with a family of injections

$$f_j : U_j \hookrightarrow A$$

such that  $A$  is the union of the  $f_j(U_j)$ .

The  $U_j$  being fixed, a gluing datum on the  $U_j$  is an isomorphism class of gluings. If  $r$  is a gluing datum, and  $(f_j : U_j \hookrightarrow A)$  is a representative of  $r$ , then  $A$  is determined, up to unique isomorphism, by  $r$ . This makes harmless the abuse of language, which we shall allow ourselves,

of saying “gluing” for “gluing datum”. For example, saying “the sum over gluings  $(f_j : U_j \hookrightarrow A)$  of the  $U_j$  of  $\varphi(A)$ ” means “the sum over gluing data  $r$  on the  $U_j$  of  $\varphi(A_r)$ , for  $(f_j : U_j \hookrightarrow A_r)$  a representative of  $r$ ”.

A gluing datum on the  $U_j$  identifies with an equivalence relation  $R$  on the disjoint sum of the  $U_j$  which induces the discrete equivalence  $x = y$  on each  $U_j$ : to  $R$ , attach the maps

$$U_j \hookrightarrow (\coprod U_j)/R.$$

For  $J = \{1, 2\}$ , it also identifies with the data of subsets  $U'_j \subset U_j$  and of a bijection

$$\varphi : U'_1 \longrightarrow U'_2 :$$

attach to it the maps  $U_j \hookrightarrow U_1 \coprod_{\varphi} U_2$ .

For  $K \subset J$  and  $(f_j : U_j \hookrightarrow A)$  a gluing of the  $U_j$ , the induced gluing of the  $U_j$ ,  $j \in K$ , is the family

$$f_j : U_j \hookrightarrow \bigcup_{j \in K} f_j(U_j).$$

**1.6.** Let  $A$  be a commutative ring. A category  $\mathcal{A}$  is said to be  $A$ -linear if the Hom-groups are endowed with  $A$ -module structures for which composition is bilinear. For  $B$  a commutative  $A$ -algebra,  $\mathcal{A} \otimes_A B$  is then the category deduced from  $\mathcal{A}$  by extension of scalars from  $A$  to  $B$ : the objects are the same, and

$$\mathrm{Hom}(X, Y) = \mathrm{Hom}_A(X, Y) \otimes_A B.$$

**1.7.** Let  $\mathcal{A}$  be  $A$ -linear. The additive envelope  $\mathcal{A}^{\mathrm{add}}$  of  $\mathcal{A}$  is the additive  $A$ -linear category of finite formal sums of objects of  $\mathcal{A}$ :

$$\begin{array}{ll} \text{objects:} & \text{finite families } (A_j)_{j \in J} \text{ of objects of } \mathcal{A}; \\ \text{morphisms } (A_j)_{j \in J} \rightarrow (B_k)_{k \in K} : & \text{a matrix } (f_k^j) \text{ of morphisms } A_j \rightarrow B_k; \\ \text{composition:} & \text{matrix multiplication.} \end{array}$$

**1.8.** A pseudo-abelian category (SGA4 I 8.7.8 says “additive and Karoubian”) is an additive category in which every idempotent endomorphism is the projection onto a direct factor. The pseudo-abelian envelope  $\mathcal{A}^{\mathrm{psab}}$  of an  $A$ -linear category  $\mathcal{A}$  is the pseudo-abelian  $A$ -linear category of formal direct factors (images of idempotent endomorphisms) of finite formal sums of objects of  $\mathcal{A}$ . See SGA4 loc. cit.

**1.9.** Let  $\mathrm{Hom}_A(\mathcal{A}_1, \mathcal{A}_2)$  denote the category of  $A$ -linear functors from  $\mathcal{A}_1$  to  $\mathcal{A}_2$ . The constructions 1.7 and 1.8 have a universal property: if  $\mathcal{B}$  is  $A$ -linear and additive (resp. pseudo-abelian), the restriction functor

$$\mathrm{Hom}_A(\mathcal{A}^{\mathrm{add}}, \mathcal{B}) \longrightarrow \mathrm{Hom}_A(\mathcal{A}, \mathcal{B})$$

(resp.

$$\mathrm{Hom}_A(\mathcal{A}^{\mathrm{psab}}, \mathcal{B}) \longrightarrow \mathrm{Hom}_A(\mathcal{A}, \mathcal{B}))$$

is an equivalence. It is “the same thing” to give an  $A$ -linear functor from  $\mathcal{A}$  to  $\mathcal{B}$ , or from  $\mathcal{A}^{\mathrm{add}}$  (resp.  $\mathcal{A}^{\mathrm{psab}}$ ) to  $\mathcal{B}$ .

An  $A$ -bilinear functor

$$\mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A}^{\mathrm{add}}$$

therefore extends to an  $A$ -bilinear functor

$$\mathcal{A}^{\text{add}} \times \mathcal{A}^{\text{add}} \longrightarrow \mathcal{A}^{\text{add}},$$

and this extension is unique up to unique isomorphism. The same holds with “add” replaced by “ps ab”.

If the  $A$ -linear category  $\mathcal{A}$  is endowed with an  $A$ -bilinear ACU tensor product (associative, commutative, with unit), then  $\mathcal{A}^{\text{add}}$  and  $\mathcal{A}^{\text{ps ab}}$  inherit it. If the tensor product of  $\mathcal{A}$  is rigid, then that of  $\mathcal{A}^{\text{add}}$  (resp.  $\mathcal{A}^{\text{ps ab}}$ ) is rigid as well.

**1.10.** A partition  $\lambda$  is a decreasing sequence of integers  $(\lambda_a)_{a \geq 1}$ , zero for  $a$  sufficiently large. Put

$$|\lambda| = \sum_a \lambda_a,$$

and say that  $\lambda$  is a partition of  $\ell$  if  $|\lambda| = \ell$ . A sequence  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r$  is viewed as a partition by prolonging it by zeros. The transposed partition  $\lambda^*$  of a partition  $\lambda$  is characterized by the equivalence between  $\lambda_i \geq j$  and  $\lambda_j^* \geq i$ . If  $\lambda$  is a partition of  $\ell$ , we shall also write  $\lambda$  for the following objects attached to  $\lambda$ .

- (A) The corresponding isomorphism class of irreducible representations of  $S_\ell$ ; its class in the Grothendieck group  $R(S_\ell)$  of representations of  $S_\ell$ , and its character. The trivial representation of  $S_\ell$  corresponds to the coarse partition  $\ell$  of  $\ell$ .
- (B) For  $N$  large enough that  $\lambda_{N+1} = 0$ : the dominant weight  $(\lambda_1, \dots, \lambda_N)$  of  $\text{GL}(N)$ .

For each  $\ell$  and each partition  $\lambda$  of  $\ell$ , assume chosen an irreducible representation of class  $\lambda$  of  $S_\ell$ . The category of sets  $U$  with  $\ell$  elements and bijections between them being equivalent to its full subcategory reduced to the object  $\{1, \dots, \ell\}$ , it is the same to give such a representation  $V_\lambda$ , or a functor  $U \mapsto V_\lambda[U]$ . In particular, the choice of  $V_\lambda$  defines, for every set  $U$  with  $\ell$  elements, a representation of  $S_U$ , which we shall again denote by  $\lambda$ . It is deduced from  $V_\lambda$  by twisting by the  $S_\ell$ -torsor of bijections from  $\{1, \dots, \ell\}$  to  $U$ .

In the preceding discussion, the base field is  $\mathbb{Q}$ . One could even take the base ring to be  $\mathbb{Z}(1/\ell!)$ . The case of other base fields or rings is obtained by extension of scalars. If  $X$  is a representation of  $S_U$ , put

$$X_\lambda := \text{Hom}(\lambda, X). \tag{1.10.1}$$

It is known that

$$\mathbb{Q}[S_U] \xrightarrow{\sim} \bigoplus_{|\lambda|=\ell} \text{End}(\lambda), \tag{1.10.2}$$

and it follows that the natural  $S_U$ -equivariant map

$$\bigoplus_{|\lambda|=\ell} X_\lambda \otimes \lambda \xrightarrow{\sim} X \tag{1.10.3}$$

is an isomorphism. This construction still makes sense for  $X$  in a pseudo-abelian  $\mathbb{Z}(1/\ell!)$ -linear category.

## 2. The category $\text{Rep}(S_T)$

**2.1.** Fix a finite set  $I$ . For  $U$  a finite set, write  $[U]_I$  for the permutation representation of  $S_I$  on the set  $\text{Inj}(U, I)$  of injections from  $U$  into  $I$ . When  $U$  (resp.  $I$ ) is the finite set  $\{1, 2, \dots, n\}$ ,  $U$  (resp.  $I$ ) is replaced by  $n$  in the notation.

For an injection  $\varphi : U \hookrightarrow V$ , composition with  $\varphi$ ,

$$\text{Inj}(V, I) \longrightarrow \text{Inj}(U, I),$$

induces

$$\text{res}_\varphi : [V]_I \longrightarrow [U]_I$$

whose transpose (1.2) is

$$\text{res}_\varphi^* : [U]_I \longrightarrow [V]_I.$$

One has

$$\text{res}_\varphi(e_g) = e_{g\varphi}, \quad \text{res}_\varphi^*(e_f) = \sum_{g\varphi=f} e_g.$$

**2.2.** Let  $C$  be a gluing of the finite sets  $U$  and  $V$ , and let  $D$  be the fibre product of  $U$  and  $V$  over  $C$ :

$$\begin{array}{ccc} D & \xrightarrow{a'} & U \\ b' \downarrow & & \downarrow a \\ V & \xrightarrow{b} & C. \end{array} \quad (2.2.1)$$

If  $U' := a'(D)$ ,  $V' := b'(D)$ , and if  $\varphi$  is the bijection  $b'a'^{-1} : U' \xrightarrow{\sim} V'$ , then

$$U \coprod_{\varphi} V \xrightarrow{\sim} C.$$

Put

$$\{C\}_I := \text{res}_{b'}^* \text{res}_{a'} \quad (\text{resp. } (C)_I := \text{res}_b \text{res}_a^*) : [U]_I \longrightarrow [V]_I.$$

These morphisms will also be denoted  $\{\varphi\}_I$  and  $(\varphi)_I$ , and, when this creates no ambiguity, the index  $I$  will be omitted.

Define an ordering on gluing data of  $U$  and  $V$  by saying that  $A$  is coarser than  $C$  ( $A \leq C$ ) if, with  $D$  and  $\varphi$  as above, the diagram

$$\begin{array}{ccc} D & \longrightarrow & U \\ \downarrow & & \downarrow \\ V & \longrightarrow & A \end{array}$$

is commutative; equivalently,  $A$  is defined by a bijection

$$\psi : U'' \xrightarrow{\sim} V'',$$

with  $U' \subset U'' \subset U$  and  $V' \subset V'' \subset V$ , extending  $\varphi$ .

For  $f \in \text{Inj}(U, I)$ , the image of  $e_f$  under  $\{C\}$  (resp. under  $(C)$ ) is the sum of the  $e_g$ , for  $g \in \text{Inj}(V, I)$ , such that  $f$  and  $g$  glue to a map (resp. an injective map) from  $C$  to  $I$ . This description shows that

$$\{C\} = \sum_{A \leq C} (A). \quad (2.2.2)$$

If  $A \leq C$  is defined by  $\psi : U'' \xrightarrow{\sim} V''$ , the ordered set of gluings  $A_1$  between  $A$  and  $C$  is isomorphic to the set, ordered by  $\supset$ , of subsets of  $U'' - U'$ : to  $A_1$ , defined by  $\psi_1 : U_1'' \rightarrow V_1''$ , where  $U' \subset U_1'' \subset U''$  and  $\psi_1$  is the restriction of  $\psi$ , attach  $U_1'' - U'$ . The Möbius function for the ordered set of gluing data is therefore

$$\mu(A, C) = (-1)^{|U''| - |U'|} = (-1)^{|C| - |A|},$$

and (2.2.2) is inverted as

$$(C) = \sum_{A \leq C} (-1)^{|C|-|A|} \{A\}. \quad (2.2.3)$$

### 2.3. Special cases:

- (i) If in (2.2.1)  $a'$  (resp.  $b'$ ) is bijective, so that the gluing  $C$  is defined by  $\varphi : U \hookrightarrow V$  (resp.  $\varphi : V \hookrightarrow U$ ), then

$$(C) = \{C\} = \text{res}_\varphi^* \quad (\text{resp. } (C) = \{C\} = \text{res}_\varphi).$$

- (ii) If the gluing is defined by a bijection  $\varphi$  from  $U$  to  $V$ , then  $(\varphi)$  is the functoriality of  $U \mapsto [U]_I$  for bijections, as defined by transport of structure.

**Construction (2.4).** *Let  $(U_j)_{j \in J}$  be a finite family of finite sets. The tensor product of the  $[U_j]_I$  is naturally isomorphic to the direct sum, over gluings  $(u_j : U_j \hookrightarrow C)$ , of the  $[C]_I$ .*

*Proof.* The tensor product of the  $[U]_I$  is the permutation representation defined by the  $S_I$ -set product of the  $\text{Inj}(U_j, I)$ . Each system of injections  $u_j : U_j \hookrightarrow I$  defines the gluing

$$u_j : U_j \hookrightarrow \bigcup u_j(U_j).$$

The corresponding gluing datum depends only on the  $S_I$ -orbit of  $(u_j)_{j \in J}$ , and this makes  $\coprod \text{Inj}(U_j, I)$  the disjoint sum of the  $\text{Inj}(C, I)$ , for  $(a_j : U_j \hookrightarrow C)$  a gluing datum. The conclusion follows from 1.3.  $\square$

**Examples (2.5).** (i) *The representation  $[\emptyset]_I$  is the unit representation:  $k$  with the trivial action of  $S_I$ . The corresponding morphism*

$$[U]_I \otimes [\emptyset]_I \xrightarrow{\sim} [U]_I$$

*is defined by the evident gluing of  $U$  and  $\emptyset$  into  $U$ .*

- (ii) *The algebra structure*

$$[U]_I \otimes [U]_I \longrightarrow [U]_I$$

*of  $[U]_I$  defined in 1.2 is the projection onto the direct summand corresponding to the evident gluing of  $U$  and  $U$  into  $U$ .*

- (iii) *The self-duality 1.2 of  $[U]_I$  is*

$$[U]_I \otimes [U]_I \xrightarrow{(ii)} [U]_I \xrightarrow{\text{res}_{\emptyset \rightarrow U}} [\emptyset],$$

*and the corresponding morphism  $\delta : \mathbf{1} \rightarrow [U]_I \otimes [U]_I$  is*

$$[\emptyset] \xrightarrow{\text{res}_{\emptyset \rightarrow U}^*} [U]_I \longrightarrow [U]_I \otimes [U]_I,$$

*where the second arrow is the inclusion of the direct summand  $[U]_I$ .*

**2.6.** Let  $((U_{j,k})_{j \in J_k})_{k \in K}$  be a family of families. It defines the family of the  $U_{j,k}$ , indexed by  $\coprod_k J_k$ . There is a natural bijection between:

- (A) gluing data  $C$  on the  $U_{j,k}$ ;
- (B) systems formed by:

- (a) for each  $k$ , a gluing datum  $C_k$  on the  $U_{j,k}$ ,  $j \in J_k$ ;
- (b) a gluing datum  $D$  on the  $C_k$ ,  $k \in K$ .

To  $C$ , one attaches the induced gluings, and their gluing into  $C$ .

We leave to the reader the verification that the isomorphism

$$\bigotimes_{k \in K} \bigotimes_{j \in J_k} [U_{j,k}]_I = \bigotimes [U_{j,k}]_I$$

becomes, via the isomorphisms 2.4, the isomorphism of

$$\bigoplus_{(C_k)} \bigotimes [C_k]_I = \bigoplus_{(D, C_k)} [D]_I$$

with the sum of the  $[C]_I$ , defined by the bijections  $(A) \leftrightarrow (B)$ .

**2.7.** This describes the associativity and commutativity constraints and the unit of the tensor product of representations  $[U]_I$  in terms of the description 2.4 of the tensor product. For associativity

$$([U]_I \otimes [V]_I) \otimes [W]_I \longrightarrow [U]_I \otimes ([V]_I \otimes [W]_I),$$

one identifies both sides with the sum of the  $[C]_I$ , for  $C$  a gluing of  $U, V, W$ . For commutativity

$$[U]_I \otimes [V]_I \longrightarrow [V]_I \otimes [U]_I,$$

use the fact that a gluing of  $U$  and  $V$  identifies with a gluing of  $V$  and  $U$ . The unit is  $[\emptyset]_I$ .

**Proposition (2.8).** *Let  $(U_j)_{j \in J}$  and  $(V_j)_{j \in J}$  be two families indexed by  $J$ , and, for each  $j$ , let  $U_j, V_j \hookrightarrow C_j$  be a gluing of  $U_j$  and  $V_j$ . Using 2.3, identify the tensor product of the  $[U_j]$  (resp.  $[V_j]_I$ ) with the sum of the  $[A]_I$  (resp.  $[B]_I$ ), for  $A$  (resp.  $B$ ) a gluing of the  $U_j$  (resp.  $V_j$ ). Then the matrix coefficient*

$$[A]_I \longrightarrow \bigotimes [U_j]_I \longrightarrow \bigotimes [V_j]_I \longrightarrow [B]_I$$

of

$$\bigotimes (C_j) : \bigotimes [U_j]_I \longrightarrow \bigotimes [V_j]_I$$

is the sum of the

$$(C) : [A]_I \longrightarrow [B]_I,$$

where  $C$  ranges over gluing data on  $A$  and  $B$  which, viewed by 2.6 as gluing data on the  $U_j$  and  $V_j$ , induce for each  $j$  the gluing datum  $C_j$  of  $U_j$  and  $V_j$ .

The verification, immediate from the description 2.2 of the  $(C_j)$ , is left to the reader.

**Proposition (2.9).** (i) *The morphisms  $(C) : [U]_I \rightarrow [V]_I$ , where  $U, V \hookrightarrow C$  is a gluing of  $U$  and  $V$  such that  $|C| \leq |I|$ , form a basis of  $\text{Hom}_{S_I}([U]_I, [V]_I)$ .*

(ii) *The condition  $|C| \leq |I|$  is satisfied by every gluing  $U, V \hookrightarrow C$  if  $|U| + |V| \leq |I|$ .*

*Proof.* By self-duality of the permutation representation  $[U]_I$ , the group  $\text{Hom}_{S_I}([U]_I, [V]_I)$  identifies with the space of  $S_I$ -invariants in  $[U]_I \otimes [V]_I$ , which is the permutation representation defined by  $\text{Inj}(U, I) \times \text{Inj}(V, I)$ . The  $S_I$ -invariants have a basis indexed by the  $S_I$ -orbits: to an orbit  $K$  corresponds the sum of the  $e_k$ , for  $k \in K$ . Injections  $u : U \hookrightarrow I$  and  $v : V \hookrightarrow I$  define the gluing datum

$$U, V \hookrightarrow u(U) \cup v(V).$$

Denote it  $C_{u,v}$ . It depends only on the orbit of  $(u, v)$ , and detects it. We leave to the reader the verification that the morphism from  $[U]_I$  to  $[V]_I$  defined by the orbit of  $(u, v)$  is  $(C_{u,v})$ . The gluing data  $C$  so obtained are those for which  $|C| \leq |I|$ . For the others,  $[C] = 0$ . This proves (i), and (ii) is trivial.  $\square$



**Proposition (2.10).** *Let  $U, V, W$  be three finite sets. Let  $U, V \hookrightarrow C$  be a gluing of  $U$  and  $V$ , and let  $V, W \hookrightarrow D$  be a gluing of  $V$  and  $W$ . Then the composite morphism*

$$(D)(C) : [U]_I \longrightarrow [W]_I$$

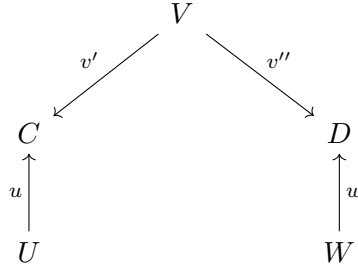
*is the sum, over gluings  $a, b, c : U, V, W \hookrightarrow A$  which induce  $C$  and  $D$ ,*

$$(D)(C) = \sum_A P_A(|I|)(A'), \quad (2.10.1)$$

*where  $A'$  is the gluing of  $U$  and  $W$  induced by  $A$ , and where  $P_A(T)$  is the polynomial with integer coefficients*

$$P_A(T) := (T - |A'|)(T - |A'| - 1) \cdots (T - |A| + 1). \quad (2.10.2)$$

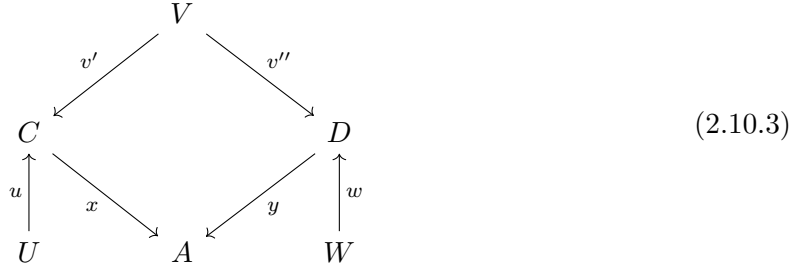
*Proof.* Write



for the gluings. By definition (2.2),

$$(D)(C) = \text{res}_w \text{res}_{v''}^* \text{res}_{v'} \text{res}_u^*.$$

The composite  $\text{res}_{v''}^* \text{res}_{v'}$  is computed by (2.2.2): it is the sum, over gluings  $x, y : C, D \hookrightarrow A$  of  $C$  and  $D$  making



commutative, of the  $\text{res}_y \text{res}_x^*$ .

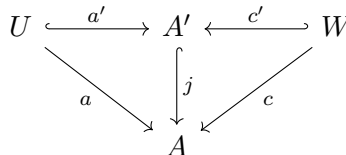
The map which sends  $x, y : C, D \hookrightarrow A$  to

$$xu, \quad xv' = yv'', \quad yw : U, V, W \hookrightarrow A$$

is bijective from the gluing data under consideration to the gluing data of  $U, V, W$  inducing the given data on  $U, V$  and  $V, W$ . For each diagram (2.10.3) one has

$$\text{res}_w \text{res}_y \text{res}_x^* \text{res}_u^* = \text{res}_{yw} \text{res}_{xu}^*,$$

and the composite  $(D)(C)$  is therefore the sum, over  $a, b, c : U, V, W \hookrightarrow A$  inducing  $C$  and  $D$ , of the  $\text{res}_c \text{res}_a^*$ . Let  $A' := a(U) \cup c(W)$ :



Then

$$\text{res}_c \text{res}_a^* = \text{res}_{c'} \text{res}_j \text{res}_j^* \text{res}_{a'}^*,$$

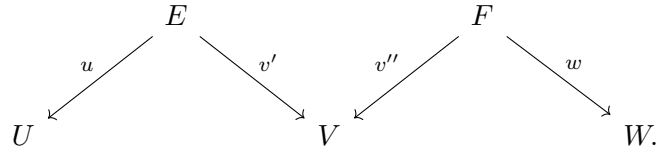
and  $\text{res}_j \text{res}_j^*$  is multiplication by the number  $P_A(|I|)$  of ways to extend an injection of  $A'$  into  $I$  to an injection of  $A$  into  $I$ . Hence

$$(D)(C) = \sum_A \text{res}_c \text{res}_a^* = \sum_A P_A(|I|) \text{res}_{c'} \text{res}_{a'}^* = \sum_A P_A(|I|)(A').$$

□

**Remark 2.11.** An analogous calculation, with (2.2.2) replaced by (2.2.3), expresses the composition of morphisms  $\{C\}$ . Assume that

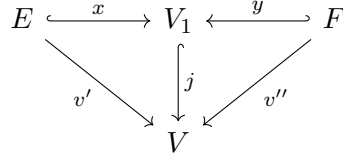
$$C = U \coprod_E V, \quad D = V \coprod_F W :$$



By definition,

$$\{D\}\{C\} = \text{res}_w^* \text{res}_{v''} \text{res}_{v'}^* \text{res}_u.$$

Put  $V_1 := v'(E) \cup v''(F)$ :



Then

$$\text{res}_{v''} \text{res}_{v'}^* = \text{res}_y \text{res}_j \text{res}_j^* \text{res}_x^* = P_{C,D}(|I|) \text{res}_y \text{res}_x^*,$$

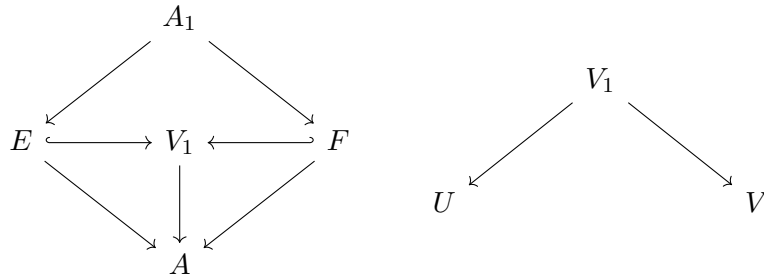
for  $P_{C,D}$  the polynomial with integer coefficients

$$P_{C,D}(T) := (T - |V_1|) \cdots (T - |V| + 1). \quad (2.11.1)$$

The composite  $\text{res}_y \text{res}_x^*$  is

$$(V_1) : [E]_I \longrightarrow [F]_I.$$

By (2.2.3), it is an alternating sum of morphisms  $\{A\}$ , for  $A$  a gluing of  $E$  and  $F$  coarser than  $V_1$ . If  $A_1$  is the fibre product  $E \times_A F$ , i.e. for  $A = E \coprod_{A_1} F$ ,



$A_1$  defines a gluing  $A'$  of  $U$  and  $V$ , and  $\{D\}\{C\}$  is the sum over  $A$

$$\{D\}\{C\} = P_{C,D}(|I|) \sum (-1)^{|V_1| - |A|} \{A'\}. \quad (2.11.2)$$